Project Title: Analysis and Prediction of Airbnb Listing Prices

BOARD INFINITY

A TRAINING REPORT

Submitted in partial fulfillment of the requirements for the award of degree of

Bachelor of Technology in Computer Science Engineering

(Data Science and Machine Learning)

Submitted To

LOVELY PROFESSIONAL UNIVERSITY

PHAGWARA, PUNJAB

From 28/05/24 to 30/06/24

SUBMITTED BY

Name of student: Perla Sai Venkata Pavan Kalyan

Registration Number: 12205536

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Student Declaration

I, Perla Sai Venkata Pavan Kalyan, bearing the registration number 12205536, hereby declare that the work done by me on "Analysis and Prediction of Airbnb Listing Prices" from May, 2024 to June, 2024, is a record of original work for the partial fulfilment of the requirements for the award of the degree, Bachelor of Technology in Computer Science and Engineering (Data Science and Machine Learning).

Perla Sai Venkata Pavan Kalyan

Dated: 30-08-2024

CERTIFICATE

BOARD

CERTIFICATE OF COMPLETION

THIS CERTIFICATE IS AWARDED TO

Perla SaiVenkataPavanKalyan

for successfully completing Course in

R Programming

10-07-2024	BOARD INFINITY	BI-20240710-7414400
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ABOUT THE COMPANY

Board Infinity: A Catalyst for Career Growth Company Vision and Mission

Vision: To empower individuals to achieve their career aspirations through innovative learning solutions.

Mission: To provide accessible, high-quality, and personalized career development programs that equip learners with the skills and knowledge needed to succeed in today's competitive job market.

Origin and Growth

Board Infinity was founded with the aim of bridging the gap between academic knowledge and industry requirements. Starting as a small-scale training institute, the company quickly gained recognition for its effective teaching methodologies and commitment to student success. Over the years, Board Infinity has expanded its operations, offering a wide range of courses and programs to cater to diverse learning needs.

Various Departments and Their Functions

Board Infinity is organized into several key departments, each playing a vital role in achieving the company's goals:

- Academic Department: Responsible for designing and developing course content, ensuring it aligns with industry standards and learner objectives.

- Faculty Department: Recruits, trains, and manages a team of experienced instructors who deliver the courses.

- Student Success Department: Provides guidance, support, and mentorship to students throughout their learning journey.

- Marketing and Sales Department: Promotes Board Infinity's offerings, attracts new students, and manages the sales process.

- Technology Department: Develops and maintains the company's technology infrastructure, including the learning management system and online platforms.

- Operations Department: Oversees the day-to-day operations of the company, ensuring smooth functioning and efficient resource allocation.

Brief Description of the Work Done

During my internship at Board Infinity, I undertook a comprehensive project focused on analyzing and predicting Airbnb listing prices. The project encompassed various stages, including data acquisition, cleaning, transformation, exploratory data analysis, feature engineering, model building, and evaluation. My primary objective was to identify key factors influencing listing prices and develop predictive models that could accurately estimate these prices. The project provided valuable insights into the dynamics of Airbnb's pricing strategies and helped in understanding customer behavior in the context of the hospitality industry.

Roles:

- Conducting extensive data preprocessing, including handling missing values, detecting and treating outliers, and transforming variables.

- Performing exploratory data analysis (EDA) to uncover patterns and relationships within the dataset.

- Developing and engineering features to improve model accuracy, including calculating distances and encoding categorical variables.

- Building and evaluating predictive models using linear regression and random forest algorithms.

- Visualizing data and model performance to communicate findings effectively to stakeholders.

- Documenting the entire process, from data cleaning to model evaluation, and presenting results in a comprehensive report.

Activities/Equipment Handled

- Software and Tools: R programming language, RStudio, and packages such as dplyr, ggplot2, corrplot, and randomForest.

- Data Handling: Managed large datasets, including data cleaning, transformation, and preparation for analysis.

- Modeling Techniques: Implemented machine learning algorithms like linear regression and random forest for predictive modelling.

- Visualization: Created visualizations such as histograms, scatter plots, box plots, and correlation matrices to explore and present data insights.

Challenges Faced and How Those Were Tackled

- Data Quality Issues: The dataset contained missing values, outliers, and inconsistent data formats. I addressed these issues by carefully cleaning the data, using techniques like imputation for missing values and median replacement for outliers.

- Feature Engineering: Developing meaningful features from the raw data was challenging, especially calculating distances and encoding categorical variables. I tackled this by thoroughly researching feature engineering techniques and using relevant packages like geosphere in R.

- Model Accuracy: Achieving a high level of accuracy in predictive modeling was difficult, particularly when balancing model complexity with interpretability. I addressed this by experimenting with different models, tuning hyperparameters, and comparing results to select the best-performing model.

Learning Outcomes

- Gained hands-on experience in data preprocessing and transformation techniques, crucial for preparing data for analysis.
- Developed a strong understanding of exploratory data analysis and its importance in identifying patterns and relationships within data.
- Acquired skills in feature engineering, particularly in creating new variables that enhance model performance.
- Learned to build and evaluate predictive models, gaining insights into the strengths and limitations of different machine learning algorithms.
- Enhanced my ability to visualize data and model results effectively, which is essential for communicating insights to non-technical stakeholders.
- Improved problem-solving skills by overcoming challenges related to data quality, feature engineering, and model accuracy.

Introduction of the Project Undertaken

Understanding Airbnb Pricing in the Digital Age

In today's digital age, platforms like Airbnb have revolutionized the hospitality industry by offering diverse and unique accommodations to travellers worldwide. From luxury villas to cozy apartments, Airbnb provides a vast array of options, catering to different preferences and budgets. This democratization of the accommodation market has empowered property owners to monetize their spaces while offering travellers a more personalized and localized experience than traditional hotels.

However, determining the optimal price for an Airbnb listing is a complex challenge. Unlike hotels, which often have standardized pricing models, Airbnb hosts must consider a multitude of factors that influence the value of their property. These factors include:

1. Location: The geographic location of the property is one of the most significant determinants of price. Listings in prime urban areas, near tourist attractions, or with scenic views generally command higher prices. Additionally, proximity to public transportation, shopping districts, and dining options can further enhance the value of a listing.

2. Property Type and Amenities: The type of property whether it's an entire home, a private room, or a shared space—also plays a crucial role in pricing. Homes with premium amenities such as a pool, hot tub, high-speed Wi-Fi, or a fully-equipped kitchen tend to attract higher nightly rates. The presence of unique features, like historical architecture or eco-friendly design, can also add to a listing's appeal and pricing.

3. Seasonal Demand: Like the broader travel industry, Airbnb prices are subject to fluctuations based on seasonal demand. During peak travel seasons, such as summer holidays, festivals, or major events, prices can increase significantly due to higher demand. Conversely, during off-peak periods, hosts may need to lower prices to attract bookings.

4. Market Competition: The pricing of nearby listings with similar attributes can impact a host's pricing strategy. Understanding the competitive landscape is essential for setting a price that is attractive to guests while still ensuring profitability for the host.

5. Guest Reviews and Ratings: A listing's reputation, as reflected in guest reviews and ratings, is another critical factor in determining price. Listings with consistently high ratings and positive reviews tend to justify higher prices, as they build trust with potential guests.

6. Occupancy Rate: Hosts must also consider the occupancy rate of their property when setting prices. Lower occupancy rates may prompt hosts to reduce prices to encourage bookings, while high occupancy rates might allow them to raise prices.

Understanding and analysing these factors is essential for Airbnb hosts who want to set competitive and profitable prices. For guests, this knowledge helps in identifying listings that offer the best value for their money. In this project, we delve into the data behind Airbnb listings, exploring the key drivers of price and building predictive models to estimate the optimal pricing strategy. This analysis not only aids hosts in maximizing their earnings but also enhances the overall experience for guests by promoting fair and transparent pricing.

Objectives of the Work Undertaken:

Exploratory Data Analysis (EDA): To uncover patterns, trends, and relationships within the Airbnb dataset that affect the pricing of listings. This involves visualizing data distributions, identifying correlations, and understanding the influence of various features on price.

Predictive Modeling: To build and evaluate machine learning models capable of predicting the price of an Airbnb listing based on its attributes. The goal is to provide a reliable pricing tool that can assist hosts in setting competitive prices.

Feature Engineering: To create new, informative features that capture the essence of a listing's characteristics, enhancing the predictive power of the models.

Outlier Detection and Handling: To identify and handle outliers in the data, ensuring that the predictive models are not unduly influenced by extreme values.

Scope of the Work:

The scope of this project includes:

- Data cleaning and preparation, including handling missing values and outliers.
- Exploratory Data Analysis (EDA) to understand the relationships between variables.
- Feature engineering to create new variables that can enhance model performance.
- Building and evaluating predictive models, including linear regression and random forest models.

- Visualizing the results to provide insights into the factors driving Airbnb prices.

Importance and Applicability:

This project is of significant importance in the context of the sharing economy, where pricing strategies can greatly influence the success of an Airbnb listing. Accurate pricing not only attracts more guests but also maximizes the revenue for hosts. The insights and tools developed in this project can be applied by Airbnb hosts, property managers, and analysts to make informed decisions about listing prices. Additionally, the methods and models developed can be generalized to similar datasets in other regions or even other platforms within the sharing economy.

Project Modules

DATA

Dataset link:
https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data/data

Dataset Description:

The dataset used in this project contains information about Airbnb listings. It includes various attributes that describe each listing such as the listings ID, name, Host name, Host Id, room type, address, reviews, overall satisfaction, accommodates, bedrooms, bathrooms, price, last modified, latitude, longitude, location, currency, rate type.

Numeric Columns:

- Room_id:

  Id number of that particular room.

- calculated host listings count:

  Number of listings owned by the host.

- reviews per month:

  Average number of reviews received per month.

- price:

  Price of the listing per night.

- minimum nights:

  Minimum number of nights required for a stay.

- longitude:

  Longitude coordinate of the listing's location.

- latitude:

  Latitude coordinate of the listing's location.

Categorical Columns:

- room type:

  Type of room available (e.g., entire home, private room, shared room).

- neighbourhood:

  Neighbourhood where the listing is located.

- neighbourhood group:

  grouping of neighbourhoods within a city or region

- name:

Name of the listing.

- host name:

  Name of the host.

Date Columns:

- last modified:

  Date of the last review.

Identifiers:

- id:

  Unique identifier of the listing.

- host id:

  Unique identifier of the host.

Data Importing:

Read the Data:

suppressWarnings(airbnb <- read_csv('..C:/Users/Pavan/Downloads/Airbnb.csv'))

Data Cleaning and Transformation:

Use dplyr and tidyr to clean the data and prepare it for analysis. This may include handling missing values, outliers, or erroneous data. Transform the data as necessary for analysis. This may include creating new variables, recoding variables, or restructuring the data.

dim(airbnb) ---> to know the dimension of the table ie. Rows x columns

str(airbnb) ----> type of each column ie. Numeric , character.

Data Cleaning:

Storing the dataset into dataframe

airbnb <-as.data.frame(airbnb)

class(airbnb)

Searching NA values

sum(is.na(airbnb))

NA's Values:

colSums(is.na(airbnb))

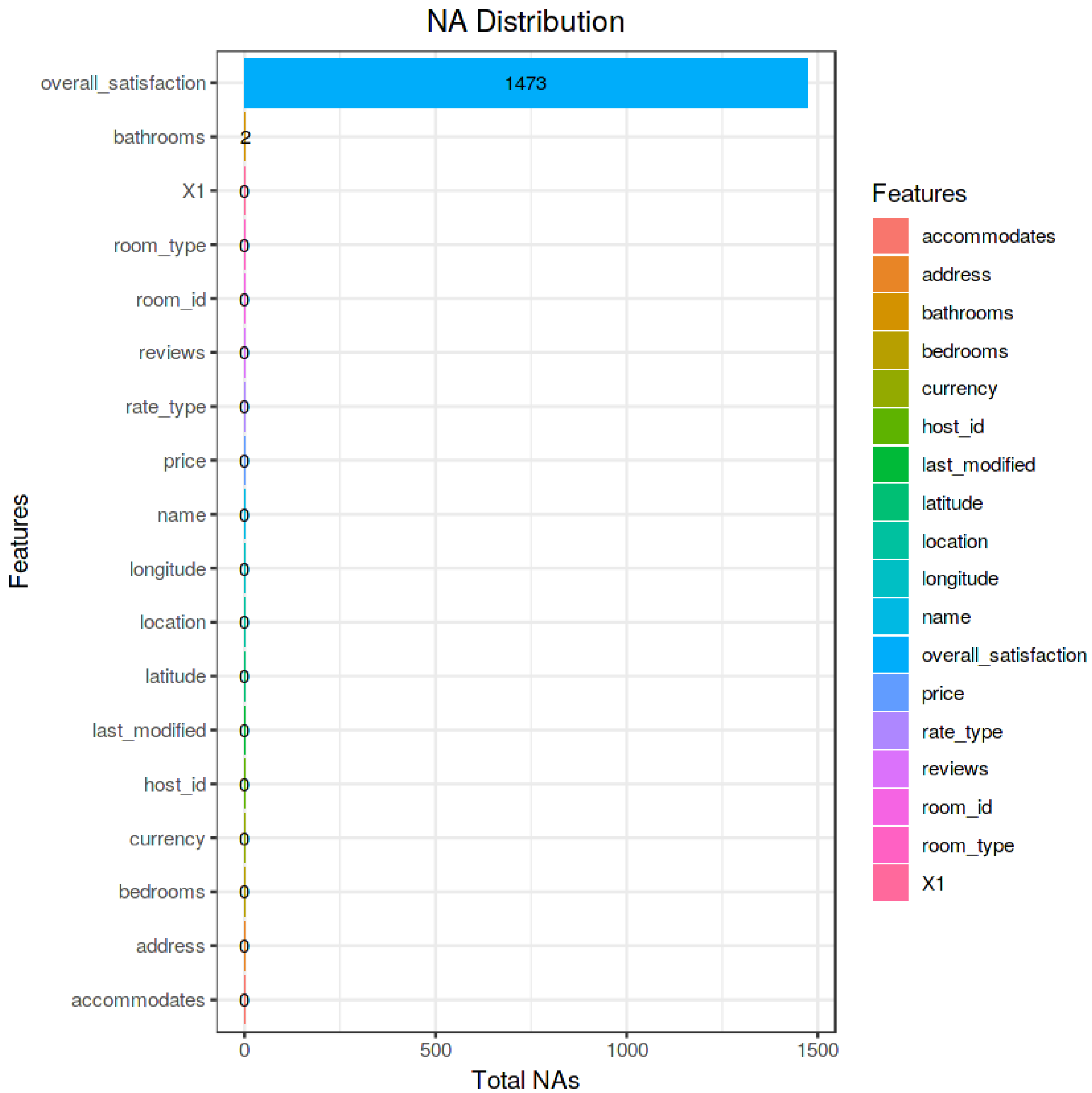
A column named “overall_satisfaction" has 1473 NAs while bathrooms has 2.

Visulaization of NA's Values:

na_vis <- data.frame(t(colSums(is.na(airbnb))))

na_bar <- data.frame(Features = names(na_vis),totals=colSums(na_vis))

```
na_bar %>% ggplot(aes(x = reorder(Features, totals), y = totals, fill = Features, label =
totals))+
  geom_bar(stat = "identity")+
  ggtitle("NA Distribution")+
  xlab("Features")+
  ylab("Total NAs")+
  coord_flip()+
  geom_text(size = 3, position = position_stack(vjust = 0.5))+
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5))
```



From the above visualisation Only overall_satisfaction & bathrooms has NA's value

NA Removal:

airbnb$overall_satisfaction[is.na(airbnb$overall_satisfaction)] <-
mean(airbnb$overall_satisfaction, na.rm = TRUE)

The mean is roughly 4.84.

```
mean(airbnb$overall_satisfaction) # it will show the mean of all values in the column named overall_satisfaction
```

```
head(airbnb$overall_satisfaction)# it shows the top five values in Overall-satisfaction
```

For the bathrooms feature, there are only 2 NAs and so we set them to zero.

```
airbnb <-airbnb %>% replace_na(list(bathrooms = 0))
```

Now confirm the absence of any NAs in the dataset:

```
sum(is.na(airbnb))
```

Feature Exploration and Selection:

```
names(airbnb) # it shows all the column names
```

There are 18 features and those less related to price prediction will be dropped to refine our EDA and Modeling Focus:

Check for unique values of the feature "address"

```
airbnb %>% select(address) %>% distinct()
```

Note that there are 27 values with different formats and 12 repeated instances of "Seattle." Any neighborhood of Seattle, the Chinese language version of "Seattle" and listings with only the State of Washington, will all be converted to Seattle. "WA" and "United States" will also be removed as they are redundant.

```
address_clean <-gsub("Seattle, WA, United States", "Seattle",
  gsub("Kirkland, WA, United States", "Kirkland",
  gsub("Bellevue, WA, United States", "Bellevue",
  gsub("Redmond, WA, United States", "Redmond",
```

```
gsub("Mercer Island, WA, United States", "Mercer Island",
gsub("Seattle, WA", "Seattle",
gsub("Renton, WA, United States", "Renton",
gsub("Ballard, Seattle, WA, United States", "Seattle",
gsub("West Seattle, WA, United States", "Seattle",
gsub("Medina, WA, United States", "Medina",
gsub("Newcastle, WA, United States", "Newcastle",
gsub("Seattle , WA, United States", "Seattle",
gsub("Ballard Seattle, WA, United States", "Seattle",
gsub("Yarrow Point, WA, United States", "Yarrow Point",
gsub("Clyde Hill, WA, United States", "Clyde Hill",
gsub("Tukwila, WA, United States", "Tukwila",
gsub("Seattle, Washington, US, WA, United States", "Seattle",
gsub("Capitol Hill, Seattle, WA, United States", "Seattle",
gsub("Kirkland , Wa, United States", "Kirkland",
gsub("Hunts Point, WA, United States", "Hunts Point",
gsub("Seattle, DC, United States", "Seattle",
gsub("Seattle, United States", "Seattle",
gsub("Vashon, WA, United States", "Vashon",
gsub("Kirkland , WA, United States", "Kirkland",
gsub("Bothell, WA, United States", "Bothell",
gsub("Washington, WA, United States", "Seattle",
    airbnb$address))))))))))))))))))))))))))
```

Replace the Chinese version of "Seattle" separately using regex:

```
address_clean2 <-gsub(".*WA*.", "Seattle", address_clean)
```

Reassign the column to the feature "address":

```
airbnb$address <-gsub("Seattle, United States", "Seattle",
                 gsub("Seattle United States", "Seattle", address_clean2))
```

Now confirm there are only 14 different cities and sort by the greatest numbers of listings:

```
city_list <-airbnb %>% group_by(address) %>% summarize(listing_sum = n()) %>%
  arrange(-listing_sum)
city_list
```

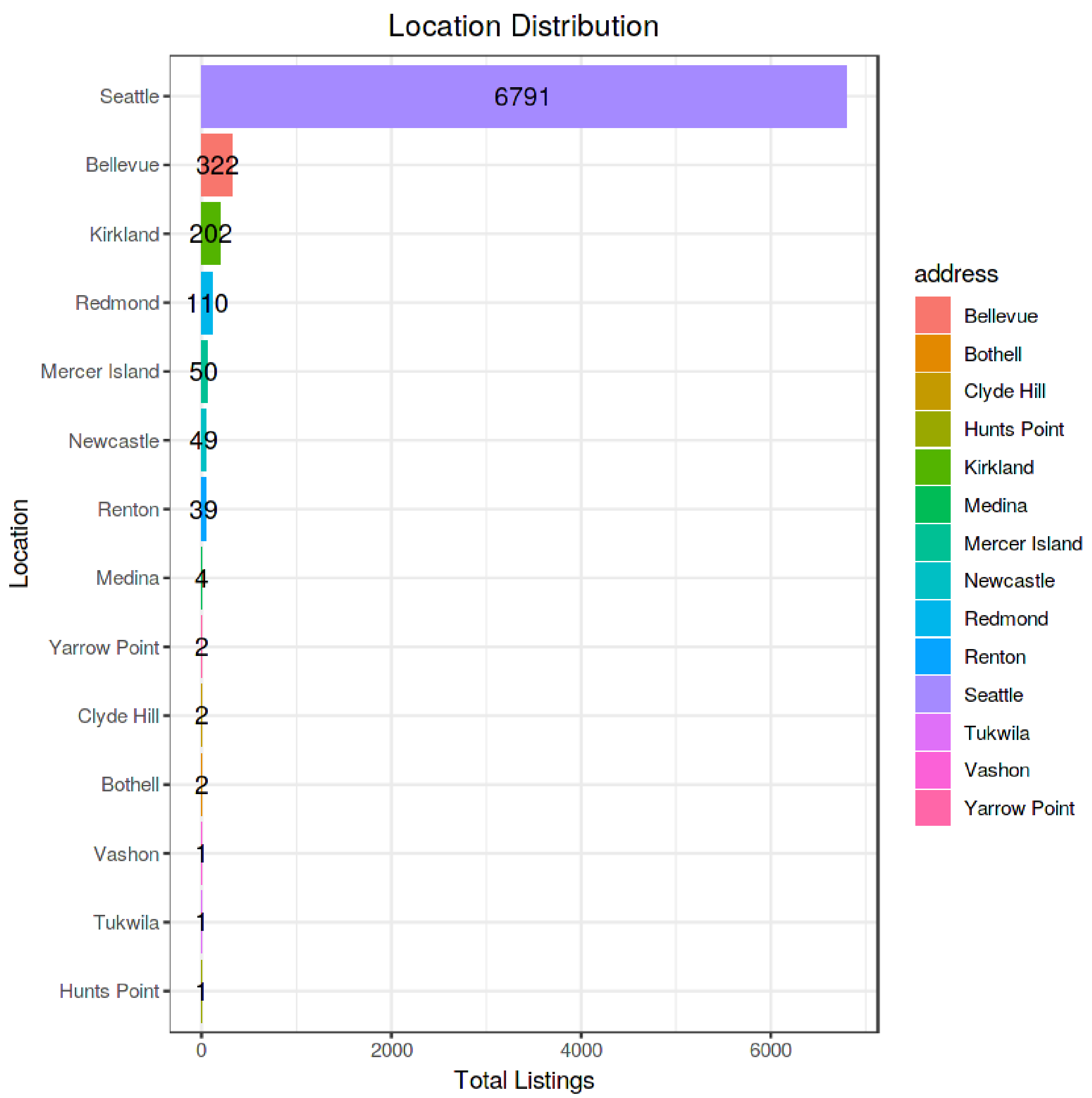
address	listing_sum
<chr>	<int>
Seattle	6791
Bellevue	322
Kirkland	202
Redmond	110
Mercer Island	50
Newcastle	49
Renton	39
Medina	4
Bothell	2
Clyde Hill	2
Yarrow Point	2
Hunts Point	1
Tukwila	1
Vashon	1

Let's explore a data visualization to confirm this:

Note: As the remaining locations are all cities, the feature "address" will later be renamed "city."

```
city_list %>%
  ggplot(aes(x = reorder(address, listing_sum), y = listing_sum,
```

```
fill = address, label = listing_sum))+
geom_bar(stat = "identity")+
ggtitle("Location Distribution")+
xlab("Location")+
ylab("Total Listings")+
coord_flip()+
geom_text(size = 4,
          position = position_stack(vjust = 0.5))+
theme_bw()+
theme(plot.title = element_text(hjust = 0.5))
```



Create the cleaned dataset:

Remove the above mentioned features and rename the columns:

```
airbnb <-airbnb %>% select(-c(X1, room_id, host_id, last_modified,
location, name, currency, rate_type)) %>% rename(city = address, rating =
overall_satisfaction, reviews_sum = reviews)
```

Reorder the columns:

```
airbnb <-airbnb[,c(8, 2, 4, 3, 1, 6, 7, 5, 9, 10)]
```

Confirm the features have been tidied and reordered with only 10 features:

```
names(airbnb)
```

1. 'price'
2. 'city'
3. 'rating'
4. 'reviews_sum'
5. 'room_type'
6. 'bedrooms'
7. 'bathrooms'
8. 'accommodates'
9. 'latitude'
10. 'longitude'

3. Exploratory Data Analysis:

- Use various R functions and packages to explore the data. This can include summary statistics, correlations, and distributions.
- Create visualizations using R's plotting capabilities. This can include scatter plots, boxplots, histograms, etc.

Correlogram:

Remove non-numeric features:

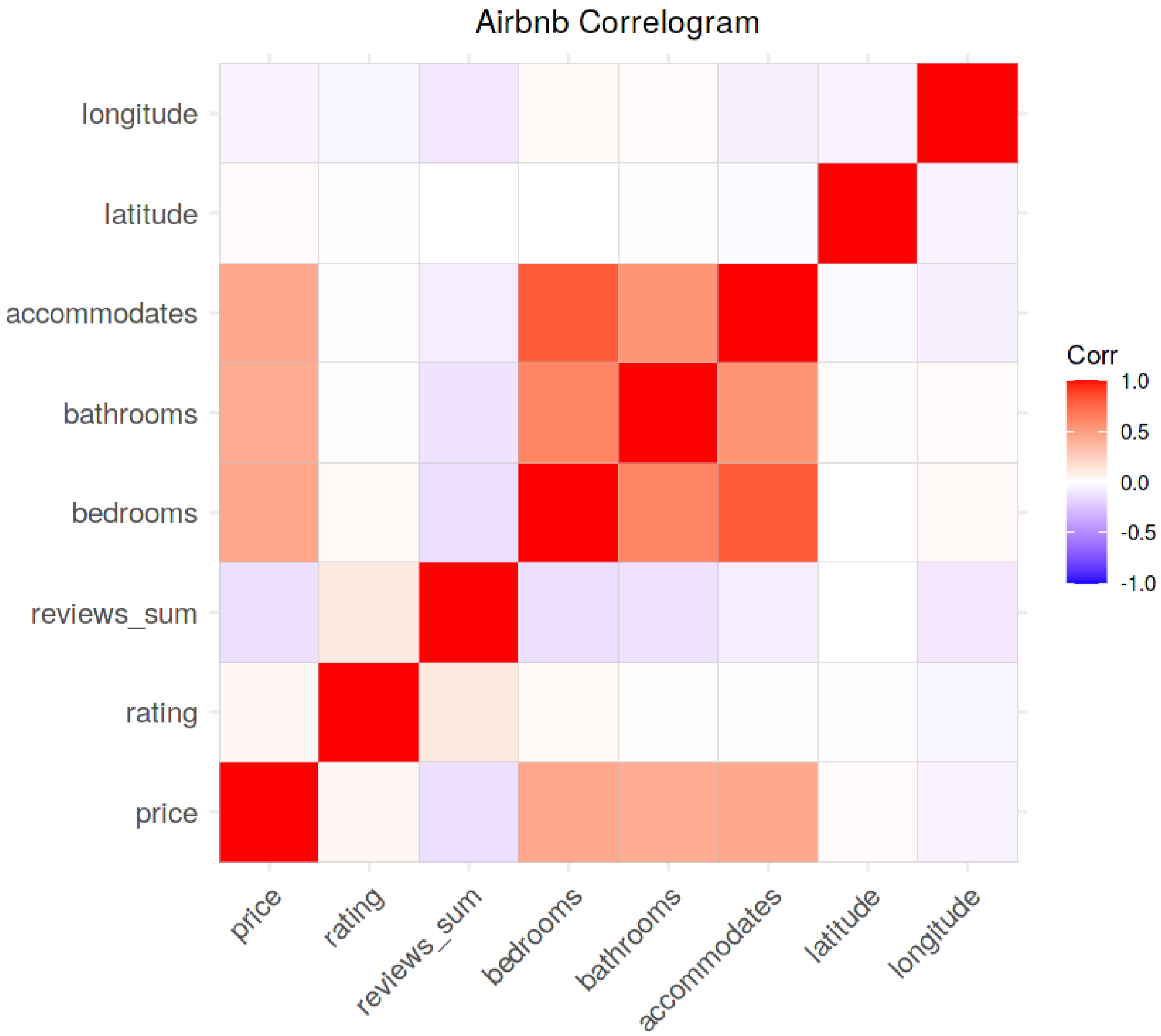
```
airbnb_num <-airbnb %>% select(-c(city, room_type))
```

Create the correlation matrix:

```
airbnb_cor <-cor(airbnb_num)
```

Plot the Correlogram:

```
ggcorrplot(airbnb_cor)+
  labs(title = "Airbnb Correlogram")+
  theme(plot.title = element_text(hjust = 0.5))
```



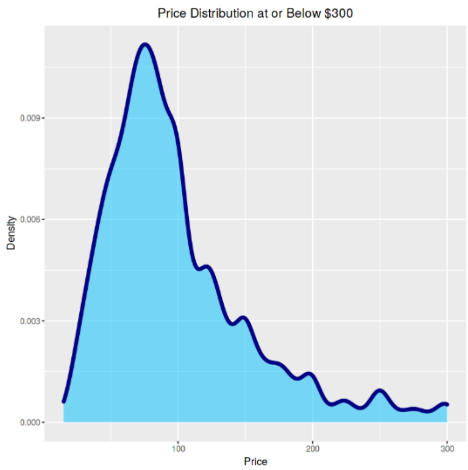
Density Plot of Price Distribution below $300:

Most common Price of bookings

---

airbnb %>% filter(price <=300) %>% ggplot(aes(price))+

```
geom_density(fill = "deepskyblue", size = 1.5, color = "navyblue", alpha = 0.5)+
xlab("Price")+
ylab("Density")+
ggtitle("Price Distribution at or Below $300")+
theme(plot.title = element_text(hjust = 0.5))
```



The most common booking price is $100/night.

Geographical Scatterplot of Prices in Seattle:

Which area are more expensive than others

```
seattle_map <- get_stamenmap(bbox = c(left = -122.5, bottom = 47.49,
                              right = -122.09, top = 47.74),
                              zoom = 9, maptype = "toner")


summary(airbnb$price)
```

Ad per visulaization Summary, we know what percentage of prices are less than or equal to 300.

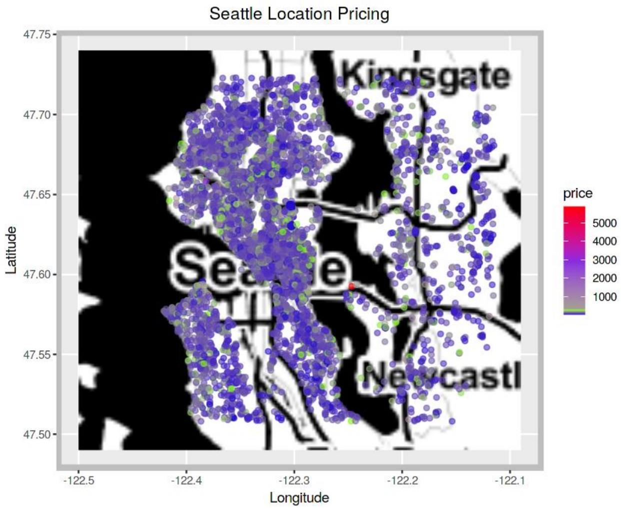
```
quantile(airbnb$price)
sum(airbnb$price <=300)/length(airbnb$price)


airbnb_map <-airbnb %>% filter(price <=300)
```

keyboard_arrow_down

Visualizing a "Heatmap" of Seattle with all listing prices included:

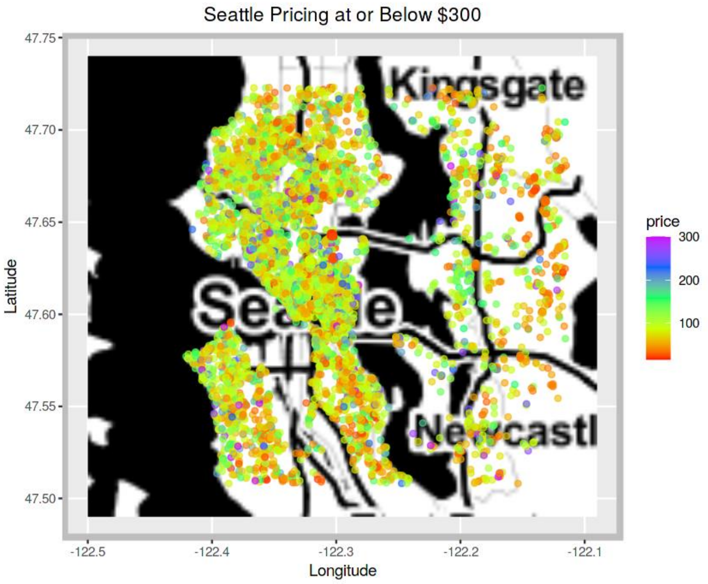
```
ggmap(seattle_map, extent= "normal")+
  geom_point(data = airbnb,
             aes(x = longitude, y = latitude, color = price),
             size = 1.5, alpha = .6)+
  scale_color_gradientn(colors = c("mediumblue", "lawngreen", "blueviolet", "red"),
             values = scales::rescale(c(.003, .013, .0176, .025, .2, .3, .4)))+
  xlab("Longitude")+
  ylab("Latitude")+
  ggtitle("Seattle Location Pricing")+
  theme(plot.title = element_text(hjust = 0.5),
        panel.border = element_rect(color = "gray", fill=NA, size=3))
```



It seems no area is significantly pricier than others, with most prices under $1000. Outliers might be skewing the data, so let's refine our heatmap by filtering it.

Heat Map of less than or equal to $300

```
ggmap(seattle_map, extent= "normal")+
  geom_point(data = airbnb_map,
             aes(x = longitude, y = latitude, color = price),
             size = 1.5, alpha = .6)+
  scale_color_gradientn(colours = rainbow(5))+
  xlab("Longitude")+
  ylab("Latitude")+
  ggtitle("Seattle Pricing at or Below $300")+
  theme(plot.title = element_text(hjust = 0.5),
      panel.border = element_rect(color = "gray", fill=NA, size=3))
```



The visualization shows most Seattle listings are between 50and150 per night, concentrated in Seattle proper. Confirm the percentage of listings in this range.

sum(airbnb$price >= 50 & airbnb$price <= 150)/length(airbnb$price)

0.699841605068638

70% of are in range from USD 50 - 150/night.

Treemap:

Sort the cities by listing_sum in a dataframe for the Treemap. Plot the Treemap to visualize the distribution of listings by city.

city_distribution <
airbnb %>% group_by(city) %>% summarize(listing_sum = n()) %>%

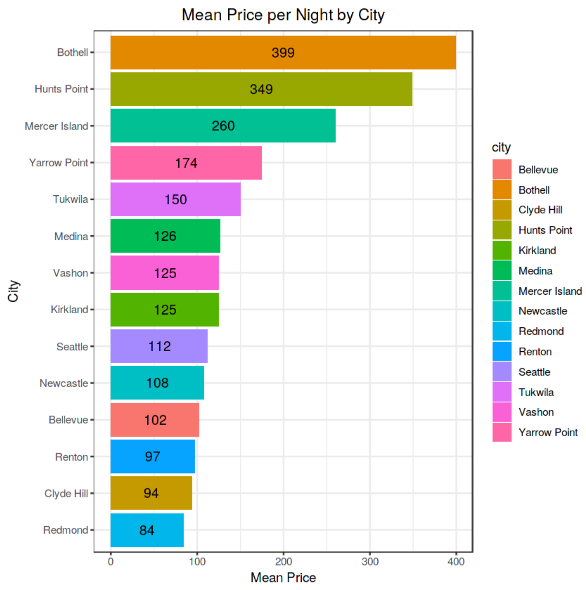
```
  arrange(-listing_sum)
# Add column "tmlab" for Treemap labels
city_distribution <-city_distribution %>%
  unite("tmlab", city:listing_sum, sep = " ", remove = FALSE)
# Plot a Treemap to visualize the distribution of listings by city:
city_distribution %>% ggplot(aes(area = listing_sum, fill = city, label = tmlab))+
  geom_treemap()+
  geom_treemap_text(fontface = "italic", col = "white", place = "center",
                    grow = TRUE)
```

Visualize Price by City:

```
city_price <-airbnb %>% group_by(city) %>%
  summarize(mean_price = mean(price),
            listing_sum = n()) %>%
  arrange(-mean_price) %>% mutate(mean_price = sprintf("%0.1f", mean_price))
# Coerce the "mean_price" to integer:
city_price$mean_price <-as.integer(city_price$mean_price)
```

Plot the Visualization:

```
city_price %>% ggplot(aes(x = reorder(city, mean_price), y = mean_price,
                          fill = city, label = mean_price))+
  geom_bar(stat = "identity")+
  coord_flip()+
  xlab("City")+
  ylab("Mean Price")+
  ggtitle("Mean Price per Night by City")+
  geom_text(size = 4,
            position = position_stack(vjust = 0.5))+
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5))
```



Visualize Mean Price & Sum of Listings by City:

Create dataframe "city_comp" with a percentage column:

city_comp <-city_price %>%

  mutate(percentage = sprintf("%0.3f",(listing_sum/sum(listing_sum)*100)))

Add the % symbol to the percentage feature:

city_comp$percentage <- paste(city_comp$percentage, "%")

Combine the mean price & percentage values into one column:

city_comp <-city_comp %>%

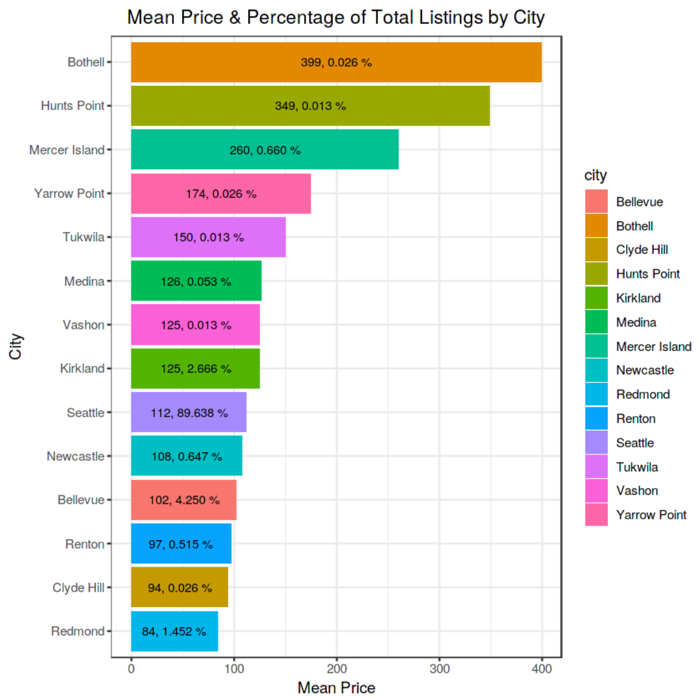
  unite("citylab", mean_price, percentage, sep = ", ", remove = FALSE)

Plot the visualization with Mean Price & Percentage of Total Listings:

city_comp %>%

  ggplot(aes(x = reorder(city, mean_price), y = mean_price,

```
                    fill = city, label = citylab))+
  geom_bar(stat = "identity")+
  coord_flip()+
  xlab("City")+
  ylab("Mean Price")+
  ggtitle("Mean Price & Percentage of Total Listings by City")+
  geom_text(size = 3, position = position_stack(vjust = 0.5))+
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5))
```



Over 89% of listings are in Seattle with an average price of 112pernight.Thetop7highestaverage−pricedlistingsmakeupabout0.8226.10 per night. This lower percentage of high-priced listings and the concentration of nearly 90% in Seattle explains the lower correlation between price and location.

4. Feature Engineering:

- Engineer new features from the existing ones that may be useful for the prediction task. For example, one might use the latitude and longitude data to create a new feature that represents the distance from a popular landmark.

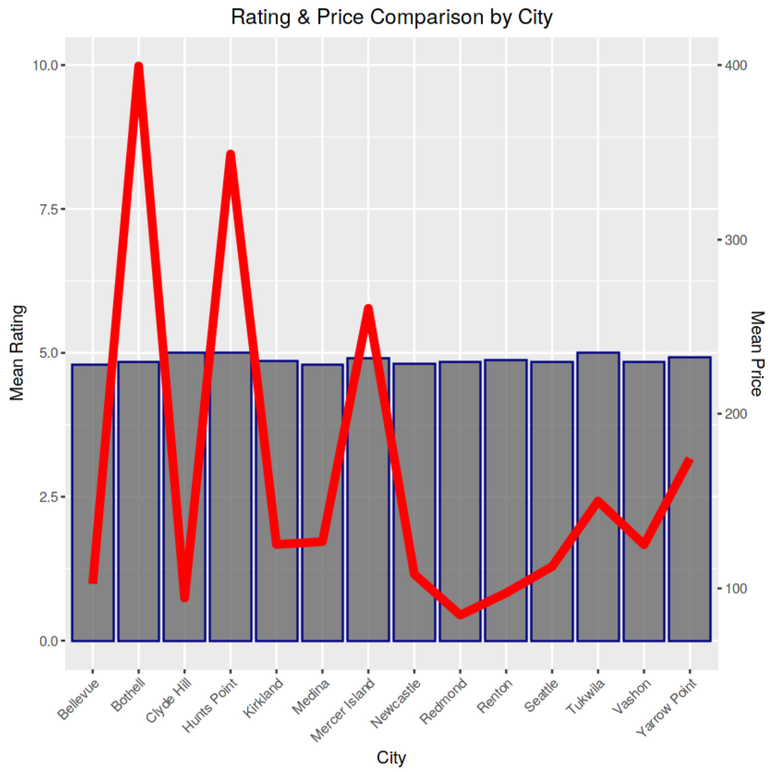
keyboard_arrow_down

Feature: "rating"

The "rating" feature ranges from 0 to 5, with a mean of 4.841. To explore its relationship with mean price by city for price prediction, create the rating_comp dataframe to compare mean price and rating. Set parameters for a dual-axis plot, then plot a bar chart for ratings with an overlapping red line for price.

```
rating_comp <-airbnb %>% group_by(city) %>%
 summarize(mean_rating = mean(rating), mean_price = mean(price)) %>%
  select(city, mean_rating, mean_price)
# Set the parameters for the dual-axis plot:
ylim_1 <-c(0,10)
ylim_2 <-c(70, 400)
b <- diff(ylim_1)/diff(ylim_2)
a <- b*(ylim_1[1] - ylim_2[1])
# Plot the Barplot (Rating) with Overlapping Line (Price):
ggplot(rating_comp, aes(city, group =1))+
  geom_bar(aes(y=mean_rating), stat="identity", color = "navyblue", alpha=.7)+
  geom_line(aes(y = a + mean_price*b), color = "red", size = 2)+
```

```
scale_y_continuous(name = "Mean Rating",
                   sec.axis = sec_axis(~ (. - a)/b, name = "Mean Price"))+
xlab("City")+
ggtitle("Rating & Price Comparison by City")+
theme(axis.text.x = element_text(angle = 45, hjust=1),
      plot.title = element_text(hjust = 0.5))
```



range(rating_comp$mean_rating)

4.79561281337047 to 5

ue to the narrow range of mean ratings by city, there's a low correlation between price and rating. For instance, high prices don't ensure higher ratings; Bothell, the most expensive city, doesn't have the highest rating. Additionally, cities with lower average prices often have higher mean ratings than Bothell

Feature: "reviews_sum"

compare the total number of reviews and different price ranges:

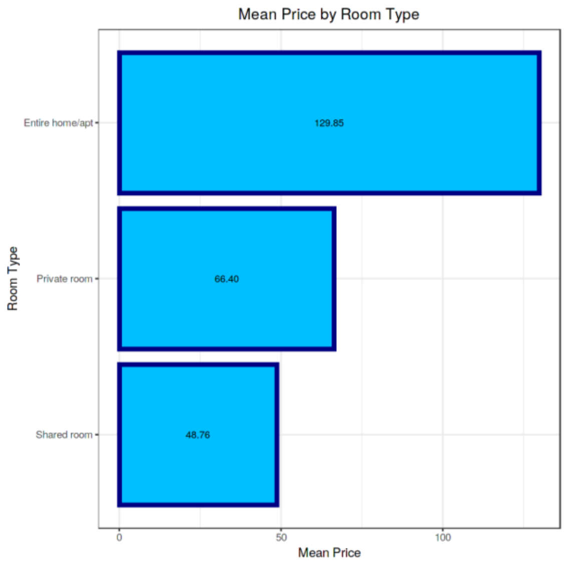
```
airbnb %>% filter(price <= 1000) %>%
  ggplot(aes(x = reviews_sum, y = price))+
  geom_point(color="navyblue", alpha = 0.6, size = 1.5)+
  xlab("Number of Reviews")+
  ylab("Price")+
  ggtitle("Review vs Price Distribution")+
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5))
```

The visualization indicates that listings with higher prices (above ~$500) generally have fewer reviews, likely due to fewer stays at these prices.

Feature: "room_type"

There are 3 room types: Entire home/apartment, Private Room, and Shared Room. Let's explore the relationship between room type and average price.

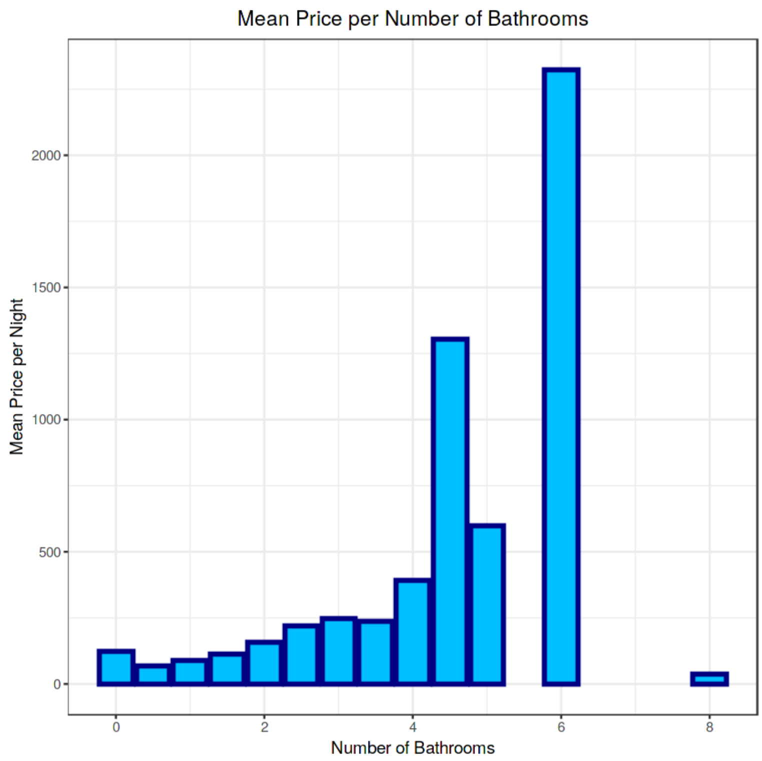
```
airbnb %>% group_by(room_type) %>%
  summarize(mean_price = mean(price)) %>%
  ggplot(aes(reorder(room_type, mean_price),
             y = mean_price, label=sprintf("%0.2f",
                                           round(mean_price, digits = 2))))+
  geom_bar(stat = "identity", color = "navyblue",
           size = 1.5, fill = "deepskyblue")+
  coord_flip()+
  xlab("Room Type")+
  ylab("Mean Price")+
  ggtitle("Mean Price by Room Type")+
  geom_text(size = 3,
            position = position_stack(vjust = 0.5))+
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5))
```



Feature: "bathrooms"

Bathrooms range from 0 to 8 in increments of 0.5 per listing. Let's create a visualization to show the mean price per listing based on the number of bathrooms available.

```
airbnb %>% group_by(bathrooms) %>% summarize(mean_price = mean(price)) %>%
  ggplot(aes(bathrooms, mean_price))+
  geom_bar(stat = "identity", fill = "deepskyblue", color = "navyblue", size = 1.2)+
  xlab("Number of Bathrooms")+
  ylab("Mean Price per Night")+
  ggtitle("Mean Price per Number of Bathrooms")+
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5))
```

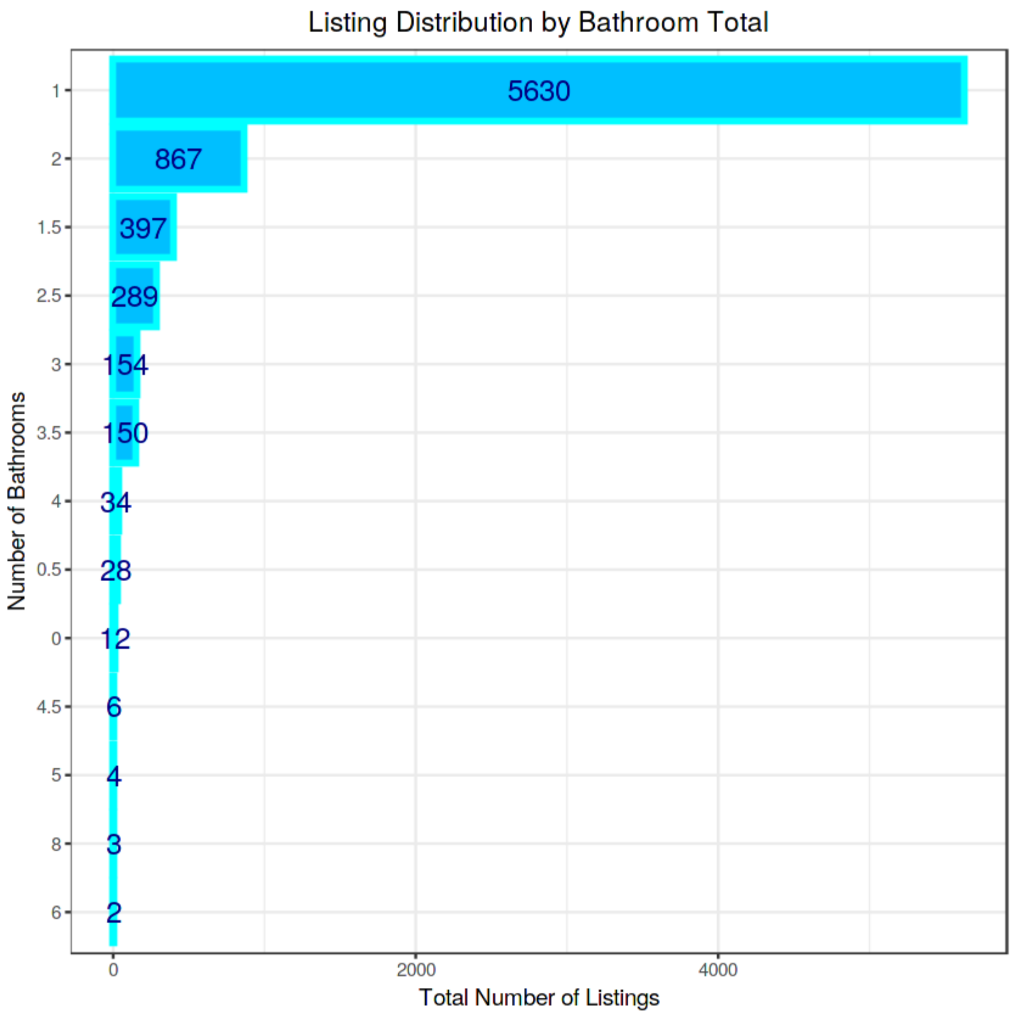


We observe that typically, higher numbers of bathrooms per listing correlate with higher prices. Interestingly, listings with 0 bathrooms had a higher average price than those with 0.5 to 1.5 bathrooms and even 8 bathrooms. This suggests that a listing with 8 bathrooms having a lower mean price than one with only one bathroom is likely an outlier.

Now, let's create a bar plot to visualize the total number of listings per bathroom count.

```
airbnb %>% group_by(bathrooms) %>% summarize(sum_bath = length(bathrooms)) %>%
%
 ggplot(aes(reorder(bathrooms, sum_bath), y=sum_bath, label = sum_bath))+
 geom_bar(stat = "identity", fill = "deepskyblue", color = "cyan", size = 1.2)+
 coord_flip()+
 geom_text(size = 5, color = "navyblue",
```

```
            position = position_stack(vjust = 0.5))+
    xlab("Number of Bathrooms")+
    ylab("Total Number of Listings")+
    ggtitle("Listing Distribution by Bathroom Total")+
    theme_bw()+
    theme(plot.title = element_text(hjust = 0.5))
```

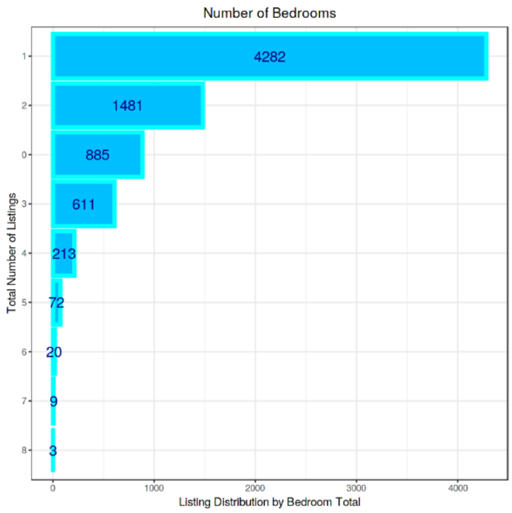


It's clear that the majority of listings, approximately 74%, have only 1 bathroom.

The visualization confirms a moderately positive correlation between price and the number of bedrooms. This feature is likely to significantly contribute to the accuracy of our price prediction models.

Now, let's explore the distribution of listings by the number of bedrooms.

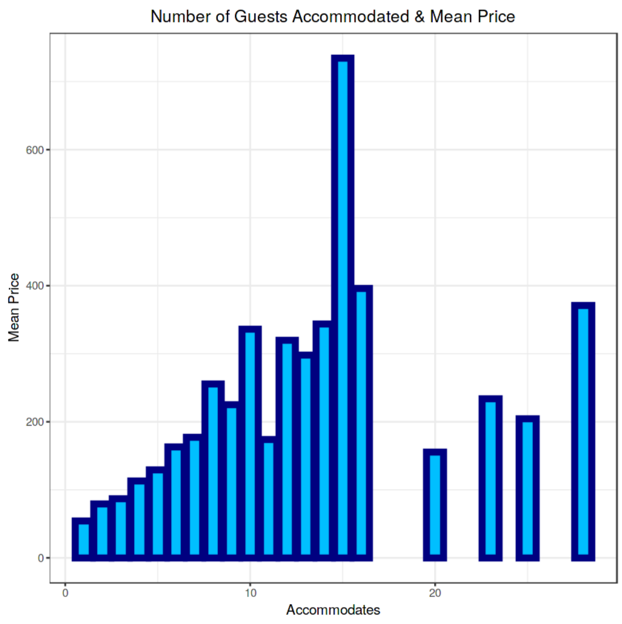
```
airbnb %>% group_by(bedrooms) %>% summarize(sum_beds = length(bedrooms)) %%
  ggplot(aes(reorder(bedrooms, sum_beds), y = sum_beds, label = sum_beds))+
  geom_bar(stat = "identity",
           color = "cyan", fill = "deepskyblue", size = 1.5)+
  coord_flip()+
  geom_text(size = 5, color = "navyblue",
            position = position_stack(vjust = 0.5))+
  xlab("Total Number of Listings")+
  ylab("Listing Distribution by Bedroom Total")+
  ggtitle("Number of Bedrooms")+
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5))
```



Feature: "accommodates"

Typically, we might predict that the mean price per night of a listing would increase with the number of guests it can accommodate. Let's visualize this relationship.

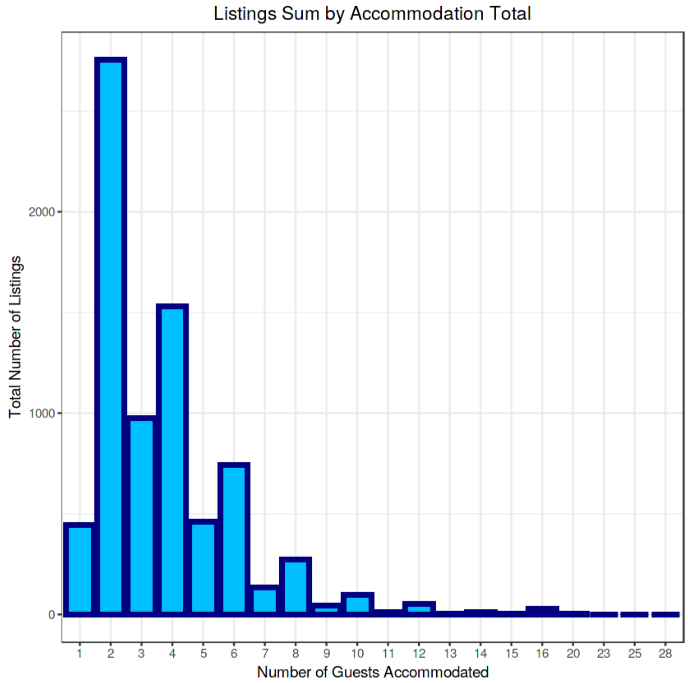
```
airbnb %>% group_by(accommodates) %>% summarize(mean_price = mean(price)) %>%
%
  ggplot(aes(accommodates, mean_price))+
  geom_bar(stat = "identity", color = "navyblue",
           size = 2, fill = "deepskyblue")+
  xlab("Accommodates")+
  ylab("Mean Price")+
  ggtitle("Number of Guests Accommodated & Mean Price")+
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5))
```



The visualization confirms that there is a general increase in average price as the number of guests a listing can accommodate increases.

Finally, let's examine the distribution of listings based on the maximum number of guests they can accommodate.

```
airbnb %>% group_by(accommodates) %>%
  summarize(sum_acc = length(accommodates)) %>%
  ggplot(aes(x = factor(accommodates), y = sum_acc))+
  geom_bar(stat = "identity", color = "navyblue", fill = "deepskyblue", size = 1.5)+
  xlab("Number of Guests Accommodated")+
  ylab("Total Number of Listings")+
  ggtitle("Listings Sum by Accommodation Total")+
  theme_bw()+
  theme(plot.title = element_text(hjust = 0.5))
```



The majority of listings can accommodate 4 or fewer guests, with 2 guests being the most popular option.

EDA Conclusion: The correlogram initially highlights significant relationships between price, bedrooms, bathrooms, and accommodation capacity, which were later confirmed through visualizations and quantitative analysis. Additionally, features like location (using latitude and longitude), rating, and review count show potential to enhance predictive model accuracy if included in our formula. "Room_type," although crucial, may need to be transformed into a quantitative scale for regression models, an aspect we will revisit in our conclusion.

5. Modeling:

- Split the data into a training set and a testing set.
- Build a regression model (or other appropriate model) to predict the price of a listing. Consider multiple different types of models, and evaluate their performance.
- Include the visualization of these models using R's capabilities.

```
# Set the seed for reproducibility:

set.seed(123, sample.kind = "Rounding")

test_index <- createDataPartition(y = airbnb$price, times = 1, p = 0.1, list = F)

airbnb_combined <- airbnb[-test_index,]

airbnb_test <- airbnb[test_index,]

# Remove test_index:

rm(test_index)
```

splits airbnb_combined into airbnb_train (80%) and airbnb_validation (20%) using a specified random seed (random_state=42) for consistency in results. Adjust the random_state parameter for different randomization

```
RMSE <-function(true_ratings, predicted_ratings){

  sqrt(mean((true_ratings - predicted_ratings)^2))

}
```

The Loss Function / RMSE

The Root Mean Square Error (RMSE) calculates the square root of the average squared differences between predicted and actual values in the test set. RMSE is preferred in regression models because it penalizes larger errors more significantly, making it a suitable metric when minimizing large errors is crucial.

Mathematically, it is defined as:

```
RMSE <-function(true_ratings, predicted_ratings){
  sqrt(mean((true_ratings - predicted_ratings)^2))
}
airbnb_train_median <-median(airbnb_train$price)
```

Conclusion of the Project

In this project, we embarked on a comprehensive analysis and prediction of Airbnb listing prices, aiming to uncover the key factors that influence the cost of accommodations. Through detailed data exploration, cleaning, and transformation, we prepared a robust dataset that allowed us to gain valuable insights into the Airbnb market.

The exploratory data analysis (EDA) revealed significant patterns and relationships between various factors, such as location, room type, and availability, that impact pricing. By employing feature engineering techniques, we enhanced the dataset with additional variables, such as distance from key landmarks and review metrics, which further contributed to the accuracy of our models.

Using R programming, we developed and compared predictive models, including linear regression and random forest models. The evaluation of these models demonstrated their ability to predict Airbnb listing prices with reasonable accuracy, providing a tool that can assist hosts in setting competitive prices and guests in finding value for their money.

Overall, this project not only highlighted the importance of data-driven decision-making in the hospitality industry but also provided actionable insights that can be leveraged by Airbnb hosts to optimize their pricing strategies. The predictive models developed serve as a foundation for further refinement and can be adapted to other regions or datasets, making them valuable assets for both current and future market analysis. Through this project, we have gained a deeper understanding of the dynamics of Airbnb pricing, contributing to more informed and strategic decisions in the sharing economy.

Future Scope

1. Enhancing Model Accuracy: Future work could focus on improving the current models by adding new features like detailed neighborhood-level data, seasonal fluctuations, and the effect of local events on pricing. Additionally, exploring advanced machine learning algorithms like gradient boosting or neural networks could lead to better predictive performance.

2. Integrating External Data: Incorporating external data sources such as weather conditions, economic indicators, or tourism statistics could provide a more comprehensive understanding of the factors that impact Airbnb pricing, leading to more accurate and context-sensitive predictions.

3. Advanced Geospatial Analysis: Extending the analysis to include the proximity of listings to public transportation, tourist attractions, and business hubs could uncover deeper insights into how location influences pricing. Using Geographic Information System (GIS) tools could further enhance spatial analysis.

4. Time-Series Forecasting: As Airbnb prices can fluctuate over time, incorporating time-series analysis would enable the prediction of future pricing trends and seasonal variations, helping hosts to optimize their pricing strategies throughout the year.

5. Personalized Pricing Recommendations: Developing models that consider user preferences and booking history could lead to personalized pricing suggestions. This approach would

allow Airbnb hosts to better cater to specific customer segments by offering tailored pricing options.

6. User-Friendly Decision Tools: The predictive model could be developed into a user-friendly application or dashboard for Airbnb hosts, enabling them to make informed pricing decisions. Such a tool could provide pricing suggestions, market alerts, and visual analytics to assist in decision-making.

7. Comparative Regional Studies: Expanding the analysis to include multiple cities or regions would allow for comparative studies, identifying common pricing factors across different markets. This broader approach could reveal insights that are not apparent when focusing on a single location.

8. Ethical Pricing Considerations: Exploring the ethical implications of pricing strategies, including the impact on affordability and accessibility, could be a valuable area of research. Future work could focus on developing fair pricing models that balance profitability with inclusivity.

9. Real-Time Dynamic Pricing Models: Implementing real-time dynamic pricing strategies that adjust based on market demand, availability, and competitor pricing could help maximize revenue while maintaining competitive pricing.

References

1. Gomonov, D. (2021). New York City Airbnb Open Data. Kaggle. Retrieved from https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data